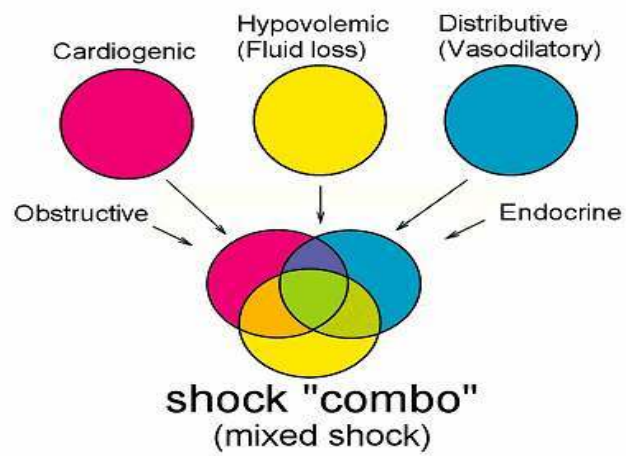


SHOCK



Shock occurs when there is an insufficient supply of nutrients, especially oxygen to the body tissues and insufficient removal of waste products.

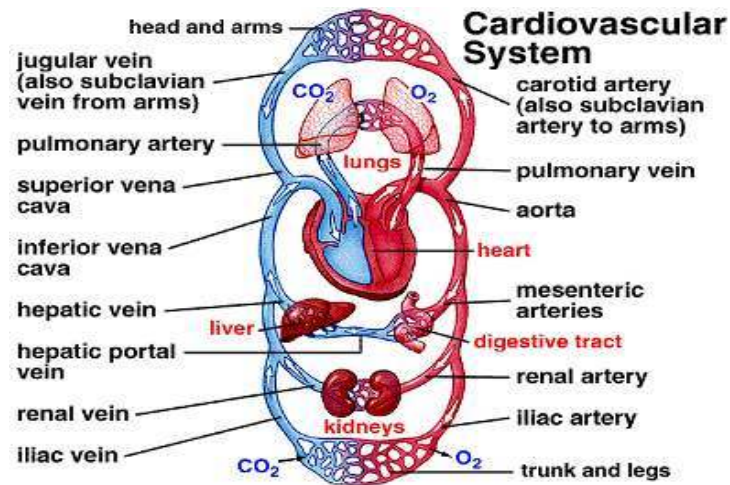
The tissues and cells of the body need a constant supply of nutrients such as oxygen, glucose and electrolytes in order to maintain their cellular activities and function, alongside this, waste products such as CO_2 and metabolites need to be removed.

If this blood supply is disrupted then cellular damage and loss of function will occur. If left untreated this lack of nutrients and build up of toxic substances will cause irreversible damage and death.

Children are very susceptible to fluid loss, dehydration and shock developing. They have higher fluid requirements per kg of body weight compared to adults. This is because they have a higher surface area to volume ratio and a higher basal metabolic rate.

For adequate perfusion to be maintained, the body needs:

- Sufficient circulating volume: Blood & Plasma
- A working vascular system: Arteries, veins & capillaries
- An effective pump: Heart (cardiac output)
- No abnormal obstruction to flow:



Insufficient tissue perfusion leading to shock can occur when there is a problem with one or all of the above components.

CLASSIFICATION OF SHOCK

Shock is classified into different categories depending on the cause of failure. However shock is a complicated clinical condition and the cause and type of cardiovascular failure often overlaps and is interrelated

Causes of shock include:

Insufficient circulating volume: Hypovolaemic shock

Vessel abnormalities: Distributive shock

Pump defects: Cardiogenic shock

Inadequate O₂ releasing capacity: Dissociative shock

Obstruction to flow: Obstructive shock

HYPOVOLAEMIC SHOCK:

The most common type of shock in paediatrics. It is frequently the end result of dehydration, leading to insufficient intravascular circulating blood volume.

Causes include:

- Diarrhea & Vomiting
- Gastroenteritis
- Burns/ scald injuries
- Haemorrhage
- Trauma
- DKA

DISTRIBUTIVE SHOCK:

Occurs as a result of uneven distribution of blood flow to the tissues.

There is a mismatch of blood delivered to the tissue beds compared with tissue requirements, as a result some tissue beds receive too much blood flow whilst others don't receive enough.

Distributive shock is most commonly caused by a severe infection such as meningococcal sepsis or an anaphylactic reaction, also known as septic or anaphylactic shock.

Overwhelming sepsis or anaphylaxis activates the release of endotoxins (sepsis) or histamine and bradykinin (anaphylaxis). The body tries to fight these harmful toxins through its inflammatory response, causing vasodilation and capillary leak in some tissues and vasoconstriction in others.

This mismatch to blood flow and tissue requirements causes hypovolaemia and acidosis to develop due to the low blood volume in the intravascular space and blood pooling in the extravascular space as capillary permeability increases.

Causes include:

- Septicaemia (septic shock)
- Anaphylaxis (anaphylactic shock)
- Vasodilating drugs

- Anaesthetic drugs
- Spinal cord injury (neurogenic shock)

CARDIOGENIC SHOCK:

The intravascular circulating blood volume is adequate but cardiac output is low due to myocardial dysfunction (arrhythmia's or dysfunction of the myocardium).

Causes include

- Severe Hypoxia
- Arrhythmias (SVT, bradycardia)
- Cardiomyopathy
- Congenital heart disease
- Drug overdose
- Hypothermia
- Severe acid base/ electrolyte balance
- Prolonged untreated hypovolaemic shock causing ischaemic damage to the myocardium

DISSOCIATIVE SHOCK:

The circulating blood volume is adequate but the haemoglobin molecule is unable to give up the oxygen to the tissues. Tissue perfusion is adequate but oxygen release to the tissues is abnormal.

Causes include:

- Profound anaemia (sickle cell)
- Carbon monoxide poisoning
- Methaemoglobinaemia.

OBSTRUCTIVE SHOCK:

Blood flow and therefore circulation is obstructed which can cause cardiac arrest. It is uncommon in paediatrics

- Tension pneumothorax
- Haemo pneumothorax
- Flail chest
- Cardiac Tamponade
- Pulmonary embolism

STAGES OF SHOCK

Shock can be described as compensated or decompensated. It is a progressive condition which if left untreated will eventually lead to irreversible damage. It is therefore essential that recognition and treatment is initiated in the early stages of development.

Initial stage:

During this stage, low perfusion causes hypoxia. Due to this lack of oxygen, the cell membranes are damaged and become leaky to extra-cellular fluid. The lack of oxygen causes the cells to perform anaerobic respiration and a build-up of lactic acid and metabolic acidosis starts to develop. Toxin removal from the liver cells is impaired due to the lack of oxygen.

Compensated Shock.

The body uses physiological mechanisms, in an attempt to reverse the developing condition of metabolic acidosis.

Due to the acidosis the child starts to hyperventilate to get rid of CO_2 , in an attempt to raise the blood PH. The baroreceptors in the arteries detect hypotension, which activates the release of adrenaline and noradrenaline. Noradrenaline mainly causes vasoconstriction with a mild increase in heart rate, whereas adrenaline mainly increases heart rate with a small effect on blood vessel tone, the combined effect results in an increase in blood pressure.



BLOOD PRESSURE IS MAINTAINED IN COMPENSATED SHOCK

The renal system activates the Renin Angiotensin system to conserve fluid via the kidneys. These hormones also cause vasoconstriction of the kidneys, gastro intestinal tract and other organs to divert blood to the

heart, lungs and brain. The lack of blood to the renal system causes low urine production and output.

Decompensated Shock.

If the cause of shock is not successfully treated and reversed, the compensatory mechanisms begin to fail. Due to the decreased perfusion of the cells, sodium ions build up within while potassium ions leak out. Anaerobic metabolism continues and increases the body's metabolic acidosis. The prolonged vasoconstriction also causes the vital organs to be compromised due to reduced perfusion.

COMPENSATION OF BLOOD PRESSURE IS NO LONGER

MAINTAINABLE AND BLOOD PRESSURE STARTS TO FALL.

As this continues the vital organs start to fail until the shock can no longer be reversed. This is known as the refractory stage and brain damage and cell death have occurred.

CLINICAL SIGNS ASSOCIATED WITH SHOCK:

Shock is a complicated clinical condition, which affects all systems of the body. In simple terms the body cells and tissues are compromised due to a lack of oxygen and nutrients with a build up of waste products and toxins and this is reflected in the clinical signs.

Respiratory System:

- Often the child who requires PICU intervention will be intubated and ventilated to manage overall treatment.
- Depending on the type and severity of shock the airway may be maintainable or unmaintainable. The child with anaphylactic shock may show signs of airway swelling, with an audible stridor and the child who is in severe shock and unconscious will have an unmaintainable airway both requiring intubation.
- The respiratory rate initially increases to compensate for the lack of oxygen and to compensate for the metabolic acidosis by “blowing off” CO_2 . The child will have an increased respiratory rate and work of breathing.
- If the acidosis is left untreated the child will develop further respiratory compromise due to increased work of breathing, CO_2 levels and exhaustion. Their respiratory rate will become slower and more irregular until intubation and manual ventilation is required to take over the work of breathing.

NURSING OBSERVATIONS.

Airway

Respiratory noises grunting, stridor, wheeze

Respiratory rate and pattern (fast, normal, slow)

Respiratory effort (head bobbing, nasal flaring, use of accessory muscles all indicate increased work of breathing)

Colour (cyanosis and pallor)

Chest movement, chest auscultation

Oxygen or ventilation requirements

Oxygen saturation (SaO_2)

Blood gas analysis

Cardiovascular System:

- The heart rate is increased initially to meet increased metabolic demand from the tissues. However as tissue hypoxia and acidosis worsen bradycardia will develop (especially in the neonate).
- Pulse amplitude decreases as the amount of blood ejected with each contraction (stroke volume) decreases. This occurs in the peripheral pulses much more rapidly than in the central pulses. Weak central pulses are a warning sign of potentially imminent cardiorespiratory arrest.
- Mottled and cold extremities and a sluggish/ absent capillary refill time (> 2 seconds) are signs of poor perfusion. As perfusion worsens, pallor, skin temperature and capillary refill time worsen
- Blood pressure is maintained until approximately 40% of circulating blood volume is lost. Hypotension is a late sign of circulatory failure and decompensated shock in the untreated child. If shock continues, despite adequate oxygenation, ventilation and volume replacement therapy vasoactive drugs may be administered to support the child's cardiovascular system.

NURSING OBSERVATIONS.

Heart rate & rhythm (ECG monitoring)

Peripheral and central pulse pressure

CVP if in situ

Invasive and non-invasive Blood Pressure

Central and peripheral temperature and core/peripheral gradient (rectal and toe temperature)

Capillary refill time, skin colour/ blanching

Renal System:

A urinary output of under 0.5 to 1ml/kg/ hr is an indicator of poor renal perfusion. Urea and creatinine levels will rise as shock develops.

NURSING OBSERVATIONS

Measurement of all output

Nappy volumes

Catheterisation will be required in order to accurately measure urinary output.

Regular Urinalysis

Assessment of urea and electrolyte balance

Central Nervous System:

Poor perfusion to the brain is dependent on the severity and duration of hypoxia. For example an arrhythmia, which acutely stops cardiac output, will cause sudden loss of consciousness. Whereas a developing circulatory or respiratory failure as in hypovolaemic shock will cause a slower alteration to consciousness, agitation followed by lethargy weakness or coma may develop.

NURSING OBSERVATIONS

Behaviour/ recognition of parents

Responsiveness

Tone (floppy)

Pupil size & reaction

Glasgow Coma Scale

AVPU scale

GENERAL TREATMENT & MANAGEMENT

As previously discussed the child in developing shock needs urgent recognition and treatment to prevent the condition from worsening. Within the unit at RHSC we have children who have varying degrees and causes of shock.

For example the child with Diabetic Keto acidosis may present in A&E and be treated with fluid resuscitation over a prolonged, determined period to improve a metabolic acidosis from developing and worsening to the collapsed child in severe shock who will require multiple therapies and intervention such as mechanical ventilation, inotropic support and haemofiltration.

Treatment is aimed at re-establishing perfusion to the vital organs through restoring and maintaining the circulating blood volume. This then ensures that oxygenation and blood pressure are sufficient to maintain effective cardiac function and prevent further damage and complications from occurring.

Treatment always uses an ABC approach as per BLS and EPLS guidelines. Initial emergency management includes airway management and 100% oxygen therapy.

Adequate intravenous access also needs to be urgently gained to administer fluid and drug therapies.

IV ACCESS.

Over-the-needle cannulae e.g. venflons/jelco are the first choice for peripheral access to start resuscitation drugs and fluids. They may also be inserted into the femoral vein prior to insertion of central venous access. Through-the-needle catheters e.g. long lines such as the vygon leadcath allow for central venous access. However this is difficult to obtain in the child who is shut down due to low circulating blood volume.

If IV access is not secured after 90 seconds, or 3 attempts, intraosseous access should be attempted. This route can be used to give any drug or fluid, which would normally be given intravenously.



FLUIDS

IV Fluids are given to replace and maintain circulating blood volume. Crystalloid or colloid fluids can be used.

Crystalloid

Volume expansion is best achieved initially using an isotonic crystalloid solution. These tend to be readily available, and are fairly cheap and hypoallergenic.

While efficient at expanding the interstitial space, and correcting sodium deficits, crystalloid fluids are not as efficient at expanding the circulating volume as colloids. Large volumes are generally well tolerated by otherwise healthy young patients, but in the critically ill child with underlying cardiac or pulmonary disease, there is a risk of pulmonary oedema. 0.9% Sodium Chloride is the most commonly used crystalloid fluid in resuscitation, but Ringer's Lactate and Hartmann's are also suitable.

Dextrose solutions should not be used for fluid resuscitation

Large volumes of IV fluids containing dextrose are ineffective at expanding the intravascular space and may result in hyperglycaemia and a secondary osmotic diuresis.

Colloid and Blood products.

These are much more efficient at rapidly expanding the circulating volume. There is a potential risk of adverse reactions e.g. infection, poor clotting or anaphylaxis. Most commonly they are given following crystalloid fluids in the initial resuscitation period or as part of ongoing therapy depending on Hb/ clotting levels.

Colloids can be classified as follows:

Blood products

- Whole Blood
- Red Cell Concentrate(RCC)
- Plasma
- Plasma Protein Fraction(PPF)
- Plasma Protein Solution(PPS)
- 5% Albumin.

Gelatin Based products

- Gelofusin
- Haemacel.

FLUID RESUSCITATION

An initial bolus of 20ml/kg of 0.9% saline is normally the dose and fluid given during fluid resuscitation. Further boluses of up to 60-80ml/kg may be needed initially to stabilise the child. The type and volume of fluid given is dependent on the child's ongoing blood picture and the type and cause of shock. However initial emergency management focuses on restoring circulating blood volume.

VASOACTIVE TREATMENT

Treatment is used to increase heart rate, improve/ maintain BP and maintain urine output.

The goals of vasoactive drug therapy in shock are:

- Increase heart rate if bradycardic
- Increase cardiac output and contractility
- Redistribute cardiac output
- Improve vascular resistance

TREATMENT SPECIFIC TO CLASSIFICATION OF SHOCK:

Hypovolaemic shock

Treatment is aimed at restoring adequate systemic perfusion.

- Whole blood loss (Haemorrhage). May be external or internal. Bleeding must be controlled and circulating blood volume restored through IV fluid therapy.
- Patient should have at least 2 wide bore venous cannulae in situ, if not Central venous access. Initial bloods must be taken for FBC, group and cross match
- Surgical review should be considered (acute abdomen or post operative bleed)
- For large blood loss the Major Haemorrhage Policy should be followed.
- Burns, diarrhoea, vomiting, etc. should be treated with crystalloid solutions to balance the nature of the fluid lost. The Parkland formula, alongside maintenance fluid is given for the 1st 24 hours post burn injury.
- Regardless of the cause, the restoration of the circulating volume is priority. As soon as the airway is maintained and oxygen administered the next step is to commence iv replacement of fluids.

Cardiogenic Shock

Cardiogenic shock is generally associated with low cardiac output and inadequate systemic perfusion, even alongside adequate intravascular volume.

It is the myocardium that is not functioning. Management is aimed at optimizing ventricular preload (myocardial stretch prior to contraction) by manipulating intravascular fluid volume. This is done through diuretics, maintaining an accurate fluid balance and inotropic drug therapy to improve the heart's contractility and correct/ prevent hypotension. The main goals of treatment are the re-establishment of circulation to the myocardium, minimising heart muscle damage and improving the heart's effectiveness as a pump.

Distributive Shock

Treatment of septic shock focuses on aggressive treatment of the infection through IV antibiotics and supportive care. The septic child will be hypovolaemic due to vasodilatation and capillary leak. Treatment includes volume replacement, with crystalloid and/ or colloid infusions, respiratory support (mechanical ventilation) cardiovascular support to maintain BP and cardiac output through inotropic drug therapy. Coagulopathy and electrolyte imbalance need to be closely monitored and treated and renal function replacement (haemofiltration) is used to remove waste products such as potassium and urea, alongside free water from the blood if renal failure occurs.

Disseminated intravascular coagulation (DIC) occurs when abnormal blood clotting causes widespread microvascular thrombosis and consumption of clotting factors. If bleeding occurs, the derangement should be corrected with fresh frozen plasma (FFP) and platelet transfusions

Anaphylaxis is treated with adrenaline to stimulate cardiac performance and corticosteroids to reduce the inflammatory response.

Obstructive Shock

Obstructive shock is rare in paediatrics. Therapy consists of removing the obstruction

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